Homework 4 (Due: 4/3)

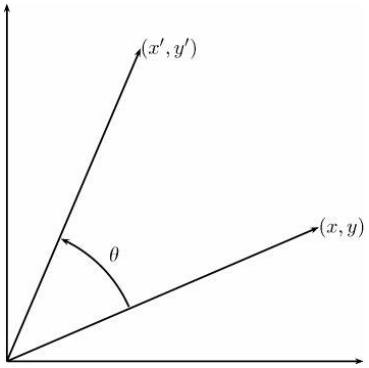
(1) Create an image consisting of a white square with a black background, e.g.,



(2) Rotate the image by 30 degrees. Use (a) rotation with nearest neighbor interpolation, and (b) rotation with bilinear interpolation.

(3) Compare the two results.

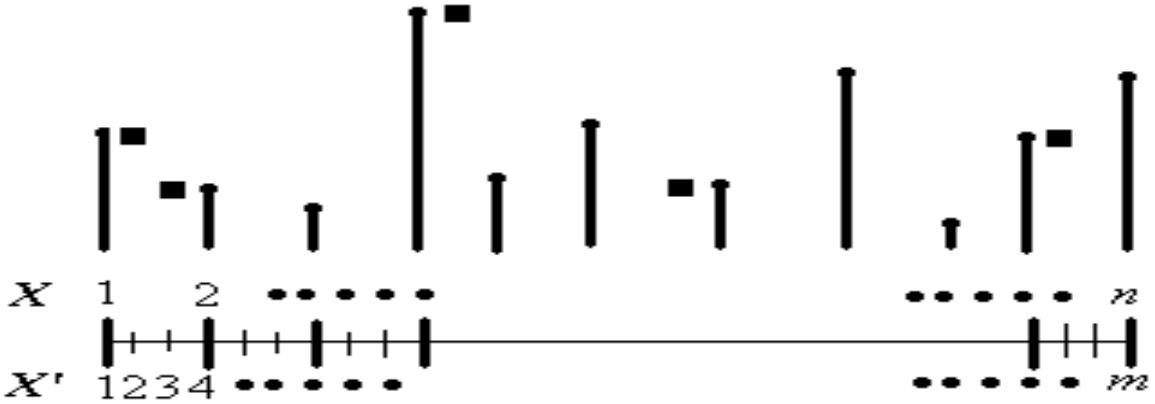
- Image Rotation



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

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- Nearest Neighbor Interpolation



- Bilinear Interpolation

